

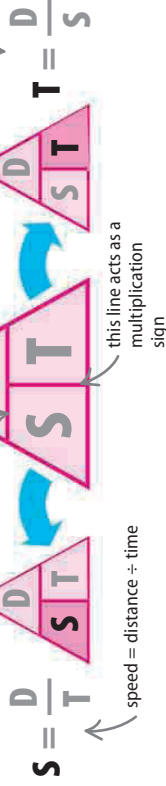
Speed

Speed measures the distance (length) traveled in a given time. This means that the formula for measuring speed is length $\div$ time. If this is measured in kilometers and hours, the unit for speed will be km/h.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

▷ Speed formula triangle

The relationships between speed, distance, and time can be shown in a triangle. The position of each unit in the triangle indicates how to use the other two measurements to calculate that unit.



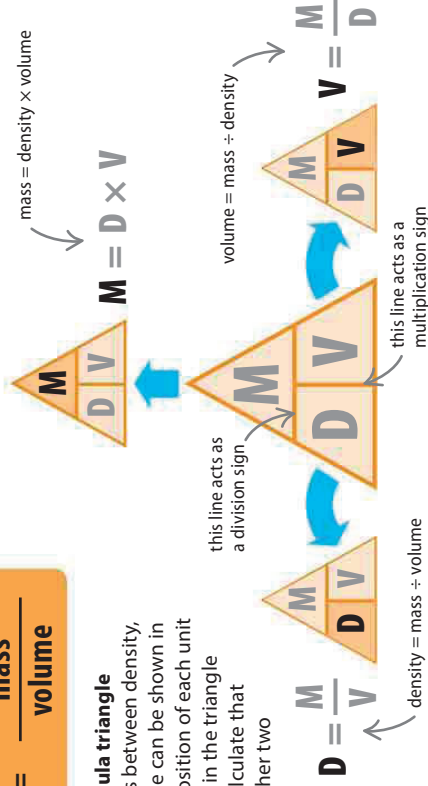
Density

Density measures how much matter is packed into a given volume of a substance. It involves two units—mass and volume. The formula for measuring density is mass $\div$ volume. If this is measured in grams and centimeters, the unit for density will be g/cm³.

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

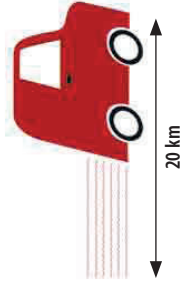
▷ Density formula triangle

The relationships between density, mass, and volume can be shown in a triangle. The position of each unit of measurement in the triangle shows how to calculate that unit using the other two measurements.



▷ Finding speed

A van travels 20 km in 20 minutes. From this information its speed in km/h can be found.



divide 20 by 60 to find its value in hours

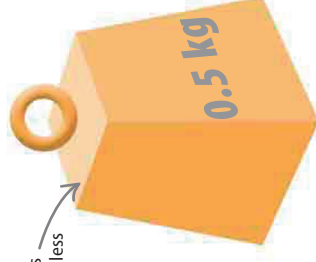
$$20 \text{ minutes} = \frac{20}{60} = \frac{1}{3} \text{ hour}$$

First, convert the minutes into hours. To convert minutes into hours, divide them by 60, then cancel the fraction—divide the top and bottom numbers by 20. This gives an answer of $\frac{1}{3}$ hour.

$$S = \frac{D}{T} = 60 \text{ km/h}$$

distance is 20 km
time is $\frac{1}{3}$ hour

Then, substitute the values for distance and time into the formula for speed. Divide the distance (20 km) by the time ($\frac{1}{3}$ hour) to find the speed, in this case 60 km/h.



density of lead is constant, regardless of mass

▷ Finding volume

Lead has a density of 0.0113 kg/cm³. With this measurement, the volume of a lead weight that has a mass of 0.5 kg can be found.

$$V = \frac{M}{D} = 44.25 \text{ cm}^3$$

mass is 0.5 kg
density is 0.0113 kg/cm³

▷ Using the formula

Substitute the values for mass and density into the formula for volume. Divide the mass (0.5 kg) by the density (0.0113 kg/cm³) to find the volume, in this case 44.25 cm³.